

7.3.4 Using power amplifiers as high current buffers in low Z VNA measurements

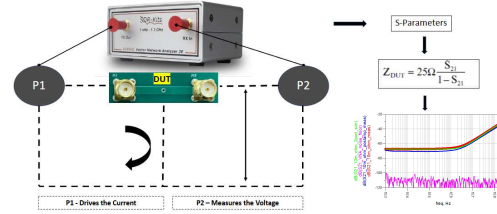
Presented by:
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05/01/2025

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Introduction

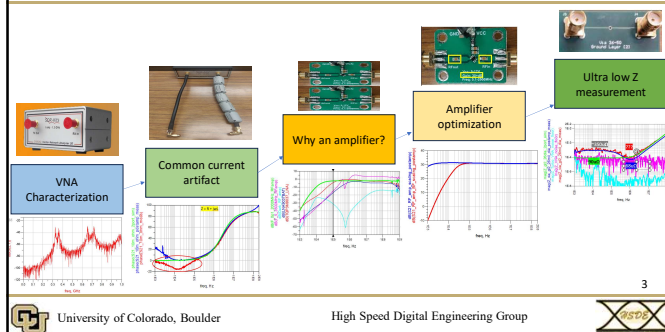
The objective of this project is to overcome limitations and measure ultra-low impedance structures, such as resistors, vias, and PDN impedance, using a low-cost SDR VNA and an RF amplifier.

Measurement setup:



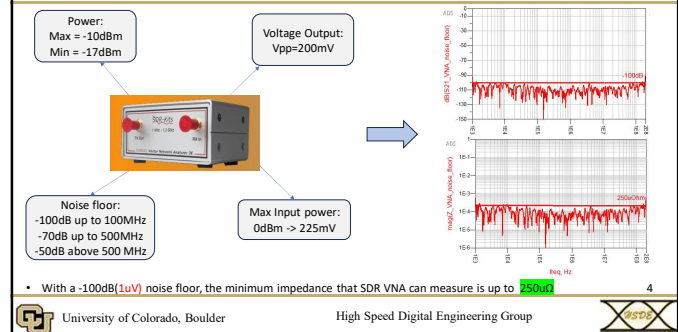
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Road map



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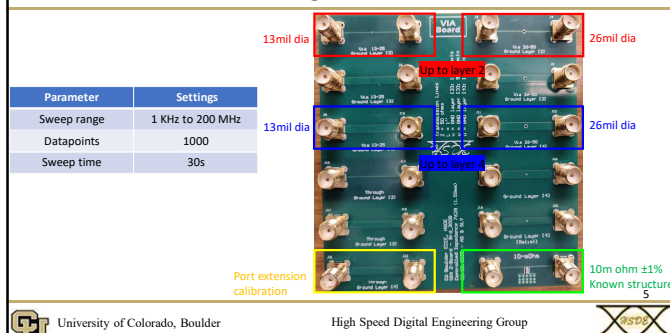
VNA Overview



• With a -100dB (1uV) noise floor, the minimum impedance that SDR VNA can measure is up to **250mΩ**

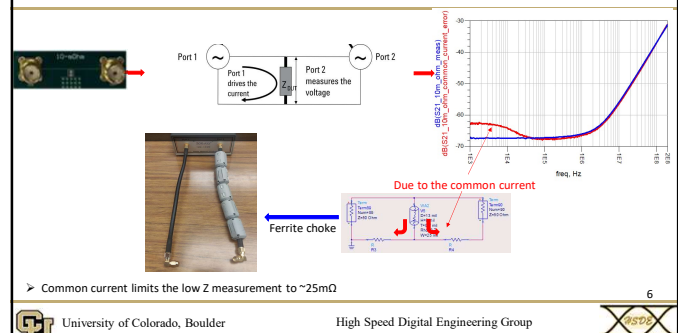
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VNA config & Test vehicle



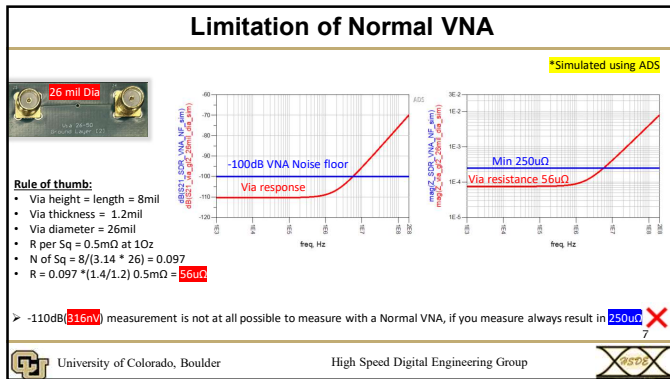
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Common current artifact

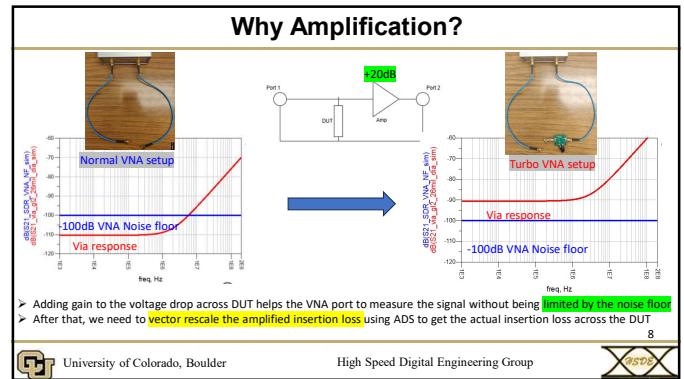


➤ Common current limits the low Z measurement to ~25mΩ

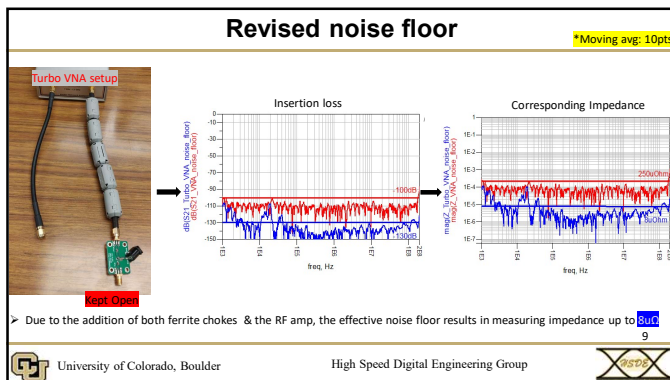
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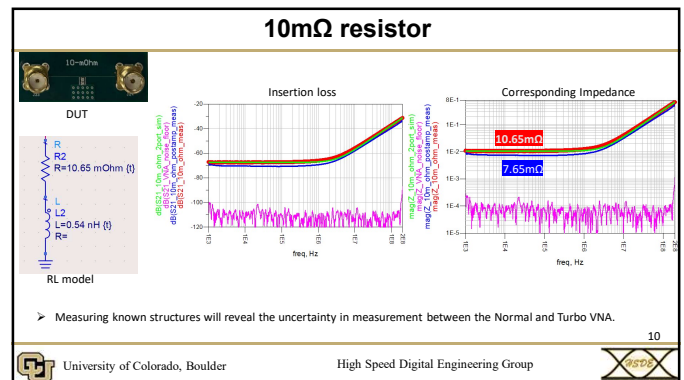
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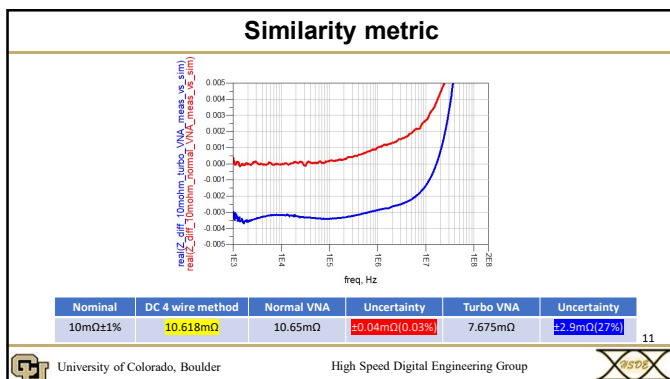
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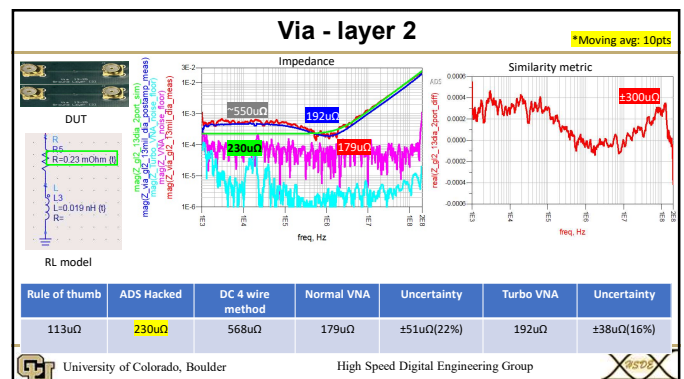
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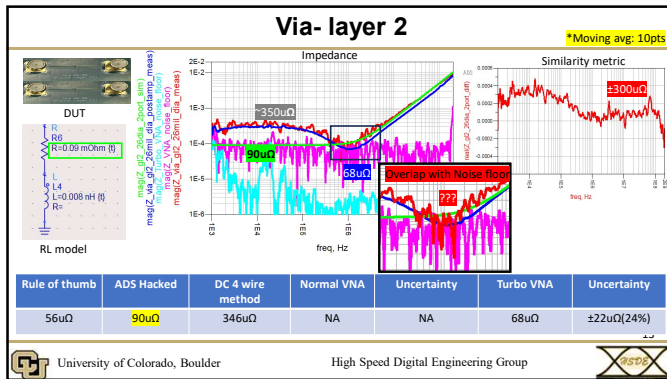
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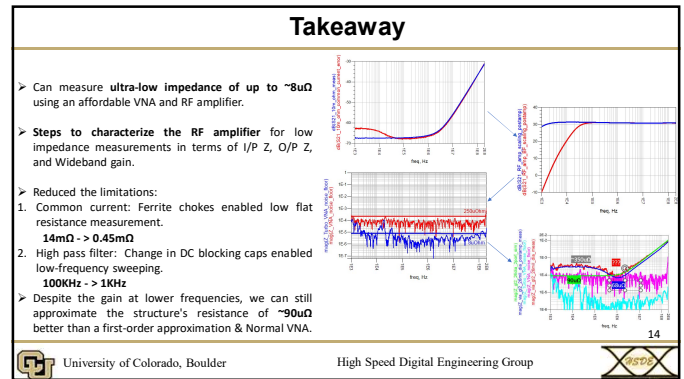
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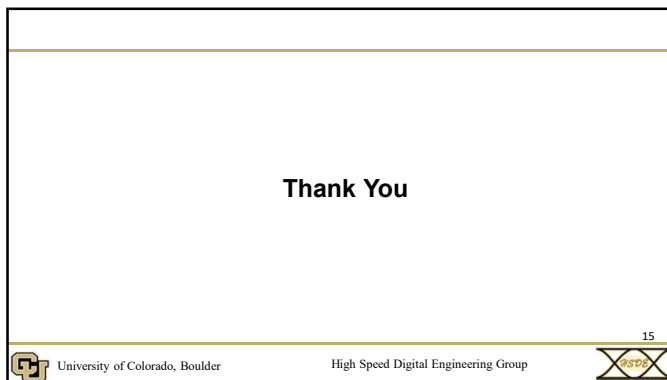
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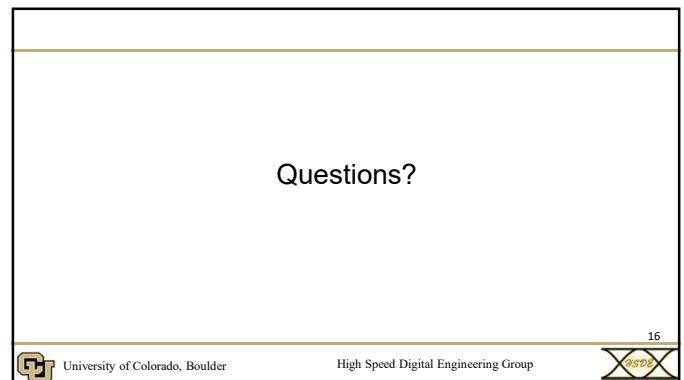
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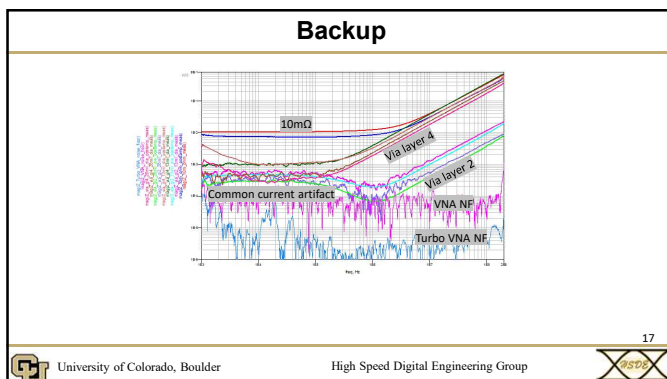
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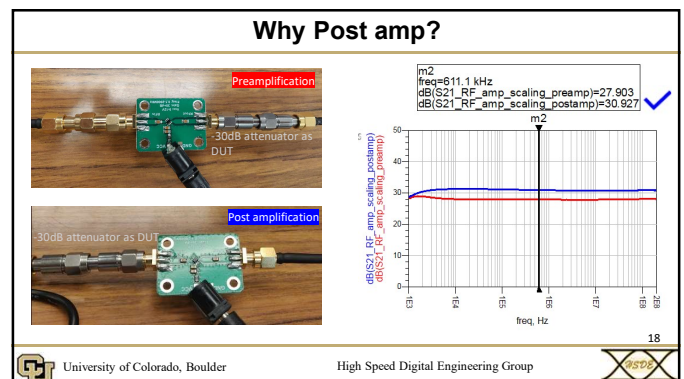
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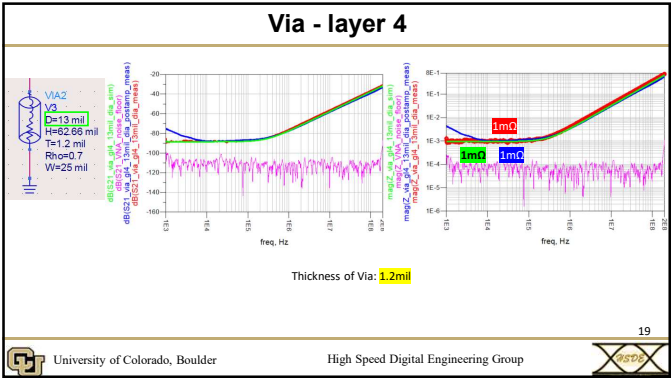
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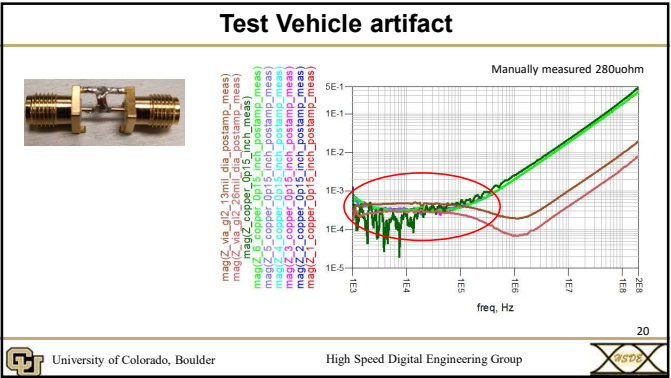
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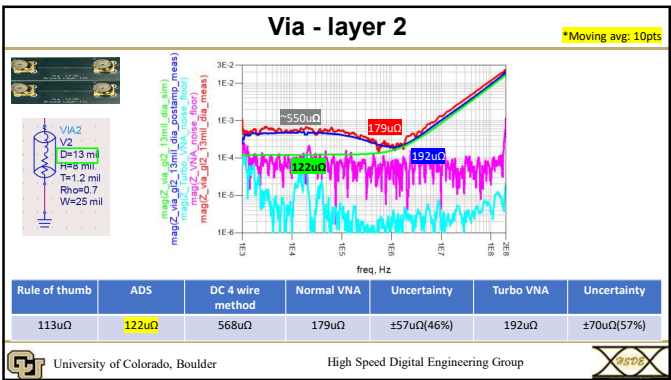
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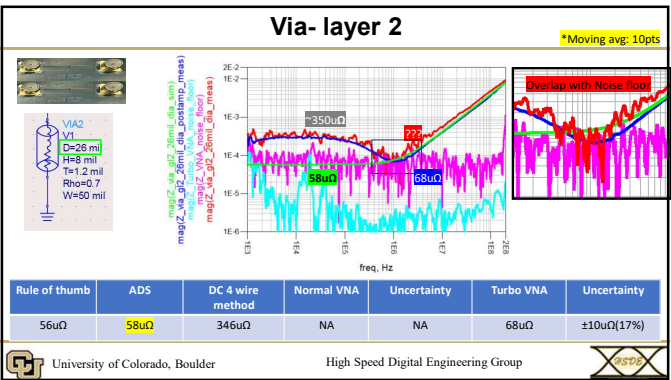
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